

Module 1: Morphology & Language Models

Linguistic Hierarchy, Tokenization (BPE), N-grams, Perplexity & Kneser–Ney Smoothing

Module I: Morphology, Tokenization & Statistical Language Modeling

1. NLP Pipeline Hierarchy (Phonetics, Morphology, Syntax, Semantics, Pragmatics)
2. Regular Expressions & Finite-State Transducers (FST) for Morphological Parsing
3. Tokenization Challenges, Minimum Edit Distance (Levenshtein) Dynamic Programming
4. Subword Tokenization Algorithms: Byte-Pair Encoding (BPE), WordPiece & Unigram
5. Word Stemming (Porter Stemmer Rules) vs. Morphological Lemmatization
6. Statistical Language Modeling & The Chain Rule of Probability
7. N -gram Language Models (Unigram, Bigram, Trigram) & Markov Assumptions
8. Language Model Evaluation: Perplexity (PPL) Mathematical Formulation
9. Smoothing Techniques: Laplace (Add-1), Add- k , Lidstone & Absolute Discounting
10. Advanced Smoothing: Good-Turing Frequency Estimation ($r^* = (r + 1) \frac{N_r + 1}{N_r}$)
11. Modified Kneser–Ney Smoothing (Interpolation & Continuation Probability P_{cont})
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 1 & 2: Natural Language Hierarchy & Morphology

Natural Language Processing (NLP) is a multidisciplinary field at the intersection of Computer Science, Artificial Intelligence, and Computational Linguistics concerned with the automated understanding, generation, and processing of human language.

Linguistic Level	Core Linguistic Phenomenon	Key Computational Tasks
1. Phonetics & Phonology	Physical sounds and linguistic acoustic patterns.	Speech-to-Text (ASR), Text-to-Speech (TTS).
2. Morphology	Structure and formation of words from root morphemes.	Stemming, Lemmatization, Tokenization, BPE.
3. Syntax	Grammatical rules governing sentence and phrase structure.	POS Tagging, Context-Free Parsing, Dependency Parsing.
4. Semantics	Literal meaning of words, phrases, and compositions.	WordNet, Word2Vec, BERT, Semantic Role Labeling.
5. Pragmatics & Discourse	Contextual meaning, intent, coreference, and conversational tone.	Anaphora Resolution, Dialogue Systems, Sarcasm Detection.

Topic 3 & 4: Subword Tokenization (Byte-Pair Encoding – BPE)

The Byte-Pair Encoding (BPE) Algorithm (Sennrich et al., 2016)

1. Initialize vocabulary with all individual characters in the training corpus plus an end-of-word symbol `.`.
2. Count the frequency of all adjacent symbol pairs in the tokenized corpus.
3. Identify the most frequent symbol pair (c_1, c_2) and merge it into a single new vocabulary token $c_1 c_2$.
4. Repeat steps 2 and 3 for a fixed number of merge operations k (e.g., 32,000 merges in GPT-2).
5. *Significance*: Solves the Out-of-Vocabulary (OOV) problem by representing rare and unseen words as sequences of subword units (e.g., `unaffordability` → `un` + `afford` + `ability`).

Topic 6 & 7: N -gram Language Modeling & The Chain Rule

1. Exact Chain Rule of Probability for Sentence $W = (w_1, w_2, \dots, w_n)$:

$$P(w_1, w_2, \dots, w_n) = \prod_{k=1}^n P(w_k \mid w_1, w_2, \dots, w_{k-1})$$

2. Bigram Language Model (1st-Order Markov Assumption):

$$P(w_1, w_2, \dots, w_n) \approx \prod_{k=1}^n P(w_k \mid w_{k-1}) \quad \text{where } P(w_k \mid w_{k-1}) = \frac{C(w_{k-1}, w_k)}{C(w_{k-1})}$$

3. Language Model Perplexity (PPL):

$$\text{PPL}(W) = P(w_1, w_2, \dots, w_n)^{-\frac{1}{n}} = \sqrt[n]{\frac{1}{P(w_1, w_2, \dots, w_n)}} = 2^{-\frac{1}{n} \sum_{i=1}^n \log_2 P(w_i \mid w_{i-1})}$$

Lower Perplexity indicates a superior language model that is less surprised by the test corpus!

Topic 9 – 11: Smoothing Techniques for Statistical Language Models

Smoothing Method	Probability Estimation Formula	Key Mechanics & Intuition
1. Laplace (Add-1)	$P_{\text{Laplace}}(w_i w_{i-1}) = \frac{C(w_{i-1}, w_i) + 1}{C(w_{i-1}) + V}$	Adds 1 to every count; shifts far too much probability mass to unseen N -grams in large vocabularies V .
2. Add- k (Lidstone)	$P_{\text{Add-}k}(w_i w_{i-1}) = \frac{C(w_{i-1}, w_i) + k}{C(w_{i-1}) + kV} \quad (0 < k < 1)$	Reduces probability mass distortion compared to Add-1.
3. Absolute Discounting	$P(w_i w_{i-1}) = \frac{\max(C(w_{i-1}, w_i) - d, 0)}{C(w_{i-1})} + \frac{d \cdot \{w : C(w_{i-1}, w) > 0\} }{C(w_{i-1})} P(w_i)$	Subtracts a fixed discount $d \in (0, 1)$ from non-zero counts and redistributes mass to unigrams.
4. Kneser-Ney (Interpolated)	$P_{\text{KN}}(w_i w_{i-1}) = \frac{\max(C(w_{i-1}, w_i) - d, 0)}{C(w_{i-1})} + \lambda(w_{i-1}) P_{\text{continuation}}(w_i)$ $P_{\text{continuation}}(w_i) = \frac{ \{w' : C(w', w_i) > 0\} }{\sum_w \{w' : C(w', w) > 0\} }$	Considers how versatile word w_i is as a continuation across diverse preceding contexts (e.g., 'San Francisco'). State-of-the-art N -gram smoothing.

Top BIT Mesra Exam Questions & Answers (Module I)

Q1. Define Perplexity. Given a test sentence with probability 10^{-6} and length 6 words, compute its Perplexity. (6 Marks)

Perplexity (PPL): The reciprocal geometric mean of the probability assigned to the test sentence by the language model:

$$\text{PPL}(W) = (P(W))^{-1/N} = (10^{-6})^{-1/6} = 10^1 = \mathbf{10}$$

A perplexity of 10 means the model is as confused as if it had to choose uniformly among 10 words at each step.